

FIELDS

Definition: A field is a set F , equipped with two operations $+$ and $\cdot$ (called addition and multiplication, respectively) obeying the following rules or axioms, $\forall x, y, z \in F$.

- (i) $x + y \in F, \quad x \cdot y \in F$
- (ii) $x + y = y + x$ (commutativity of addition)
- (iii) $(x + y) + z = x + (y + z)$ (associativity of addition)
- (iv) There is an element in F , called "zero" (denoted 0) such that $x + 0 = x$ in F .
- (v) For each $x \in F$, there is an element ^{in F .} called its "additive inverse" (denoted x_I), such that
$$x + x_I = 0$$
- (vi) $x \cdot y = y \cdot x$ (commutativity),
- (vii) $(x \cdot y) \cdot z = x \cdot (y \cdot z)$ (associativity),
- (viii) $(x + y) \cdot z = x \cdot z + y \cdot z$ (distributivity)

(ix) There is an element $1 \in F$, $1 \neq 0$, such that $x \cdot 1 = x$

(x) If $x \neq 0$, there is an element ^{in F} called multiplicative inverse of x (denoted x^{-1}) such that
$$x \cdot x^{-1} = 1$$

Example of a Field?

$\mathbb{R} = \{\text{set of all real numbers}\}$
 $+$: Regular addition of real numbers
 $\cdot$: Regular multiplication of reals.

What is 0 ? the real zero $0 \in \mathbb{R}$

What is 1 ? " " number 1 , $1 \in \mathbb{R}$

What is additive inverse of $x \in \mathbb{R}$
"negative of x "

What is the multiplicative inverse of
 $x \in \mathbb{R}$? ($x \neq 0$) $\frac{1}{x}$

SOME (OBVIOUS?) PROPERTIES OF FIELDS

① 0 is unique: Suppose there are two zeros 0_1 and 0_2 in F .

$$0_1 + x = x, \forall x \in F, \text{ choose } x = 0_2 \Rightarrow \underline{0_1 + 0_2 = 0_2}$$

Similarly, $0_2 + x = x, \forall x \in F$ choose $x = 0_1$

$$\Rightarrow \underline{0_2 + 0_1 = 0_1}$$

$$\underline{0_2 + 0_1 = 0_1} \text{ and } \underline{0_1 + 0_2 = 0_2} \quad \text{However } 0_1 + 0_2 = 0_2 + 0_1 \text{ (commutativity of addition)}$$

$$\Rightarrow \boxed{0_1 = 0_2}$$

$\Rightarrow$ There is a unique 0 in F

② 1 is unique:

Follow similar argument as above.

③ Additive inverse of every element is unique

Suppose there are two additive inverses of x , denoted x_1 and x_2 . $\Rightarrow x + x_1 = 0$ and $x + x_2 = 0$.

$$(x + x_1) + x_2 = 0 + x_2 = x_2$$

$$x + (x_1 + x_2) = x_2 \quad (\text{associativity})$$

$$x + (x_2 + x_1) = x_2 \quad (\text{commutativity})$$

$$(x + x_2) + x_1 = x_2 \quad (\text{associativity})$$

$$\underbrace{(x + x_2)}_{=0} + x_1 = x_2 \quad \Rightarrow \boxed{x_1 = x_2}$$

④ Multiplicative inverse of every (non-zero) element is unique:

Similarly establish this fact.

⑤ $a \cdot 0 = 0$:-

$$a \cdot 0 + a \cdot 0 \stackrel{\text{(Distributivity)}}{=} a \cdot (0 + 0) = a \cdot 0$$

$$\Rightarrow a \cdot 0 + a \cdot 0 = a \cdot 0$$

Let b be an additive inverse of $a \cdot 0$

(Note that b exists, by axioms of the field)

$$\Rightarrow (a \cdot 0 + a \cdot 0) + b = \underbrace{a \cdot 0 + b}_0 = 0$$

$$\Rightarrow a \cdot 0 + (\underbrace{a \cdot 0 + b}_0) = 0 \stackrel{=0}{\Rightarrow} a \cdot 0 + 0 = 0$$

⑥ $(1_I) \cdot a = a_I$:-

(Formula for computing additive inverse of any field element).

Proof:

$$a_I + a = a + a_I = 0$$

$$\Rightarrow a_I + a + 1_I \cdot a = 0 + 1_I \cdot a = 1_I \cdot a$$

$$\Rightarrow a_I + 1 \cdot a + 1_I a = 1_I \cdot a$$

$$\Rightarrow a_I + (1 + 1_I) \cdot a = 1_I \cdot a \quad \text{(Distributivity)}$$

$$\Rightarrow a_I + 0 \cdot a = 1_I \cdot a$$

$$\Rightarrow a_I + 0 = 1_I \cdot a$$

(Since $0 \cdot a = 0$)

$$\Rightarrow \boxed{a_I = 1_I \cdot a}$$

Examples of Fields

① Smallest Field?

Any field F must have a 0 and a 1 ($1 \neq 0$). Can we define $+$ and $\cdot$ so that the set $\{0, 1\}$ is a field?

$+$		
	0	1
0	0	1
1	1	0

	0	1
0	0	0
1	0	1

Additive inverse of 0 is 0
" " " 1 is 1

Multiplicative inverse of 1 is 1.

$\{0, 1\}$ is a field with $+$ and $\cdot$ defined as above.

② $\mathbb{R}$: Already shown.

③ $\mathbb{C}$: set of all complex numbers.

$$\mathbb{C} = \{ (x, y), x, y \in \mathbb{R} \}.$$

Using standard addition, multiplication of complex numbers, show $\mathbb{C}$ is a field.

$$+ : (x, y) + (p, q) = (x+p, y+q)$$

↑ ↑
real addition.

$$\begin{aligned} \cdot : (x, y) \cdot (p, q) \\ = (x \cdot p - y \cdot q, x \cdot q + y \cdot p) \end{aligned}$$

$$0 \text{ of } \mathbb{C} ? (0, 0)$$

$$1 \text{ of } \mathbb{C} ? (1, 0)$$

$$\text{additive inverse of } (x, y) ? (-x, -y)$$

multiplicative inverse

$$\text{of } (x, y) ? \text{ DIY.}$$

VECTOR SPACES

A vector space $\mathcal{V}$, defined over a field F , is a non-empty set (whose members are called vectors) equipped with two operations :-

(i) Vector addition, $+$: $\mathcal{V} \times \mathcal{V} \rightarrow \mathcal{V}$

(ii) Scalar Multiplication, $\cdot$: $F \times \mathcal{V} \rightarrow \mathcal{V}$

which must satisfy the following properties

(0) Closure under Vector Addition and Scalar Multiplication:

(follows from the way these operations are defined)

If $u, v \in \mathcal{V}$, then $u+v \in \mathcal{V}$, and $\alpha \in F, u \in \mathcal{V}, \alpha \cdot u \in \mathcal{V}$

(i) $u+v = v+u$, $\forall u, v \in \mathcal{V}$ (Commutativity of vector addition)

(ii) $(u+v)+w = u+(v+w)$, $\forall u, v, w \in \mathcal{V}$ (Associativity)

(iii) Existence of zero: There is a vector $0 \in \mathcal{V}$ such that $0+v = v$, $\forall v \in \mathcal{V}$

(iv) Existence of additive inverse: Every $v \in \mathcal{V}$ has another element $v_I \in \mathcal{V}$ such that $v+v_I = 0$

(v) $\alpha \cdot (\beta \cdot v) = (\alpha \cdot \beta) \cdot v$, $\forall \alpha, \beta \in F, v \in \mathcal{V}$ (Associativity of scalar multiplication)

(vi) $1 \cdot v = v$, $1 \in F, v \in \mathcal{V}$

(vii) $\alpha \cdot (u+v) = \alpha \cdot u + \alpha \cdot v$ (viii) $(\alpha+\beta) \cdot v = \alpha \cdot v + \beta \cdot v$
 $\forall \alpha \in F, u, v \in \mathcal{V}$ Distributivity. → $\alpha, \beta \in F, v \in \mathcal{V}$